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AI-Driven Fracture Geometry Prediction: Fast and Reliable Modeling Using XGBoost Under Reservoir Uncertainty

Lifeng Yang and Meng Fan, Research Institute of Petroleum Exploration and Development, Beijing, China; Joseph Leines Artieda, Ning Li, and Wei Yu, Beijing Karst Science & Technology LTD, Beijing, China; Yu Wang, StraightEnergy, Beijing, China; Jijun Miao, Beijing Karst Science & Technology LTD, Beijing, China

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Abstract

This paper presents an artificial intelligence (AI)-driven workflow for rapid and accurate prediction of hydraulic fracture geometry under reservoir uncertainty. Using XGBoost as a surrogate model, the method learns spatial growth patterns from 3D sampling points extracted from high-fidelity fracture simulations. This approach enables fast and reliable geometry forecasting in both lateral and vertical directions, significantly reducing the need for repeated full-physics simulations. Key geomechanical parameters were sampled using Latin Hypercube Sampling to efficiently explore a high-dimensional parameter space. A total of 100 fracture propagation simulations were run, each producing detailed time-resolved 3D fracture geometries. During simulation, structured stencils captured fracture width, pressure, and proppant concentration at defined x–y–z locations, allowing the model to learn spatio-temporal growth behavior. The trained XGBoost model was validated against hold-out simulation data and evaluated in single- and multi-cluster scenarios with varying degrees of curvature and natural fracture influence.

The surrogate model reproduced fracture length, height, and curvature trends with mean relative errors below 1%, even in complex non-planar cases. For example, in a single-cluster curved fracture case, the model achieved 0.4% length error and 0.15% height error while generating predictions for 190-time steps in 34 seconds. In multi-cluster cases, the workflow preserved fracture asymmetries and cluster interaction effects while maintaining rapid prediction times. Compared to full-physics simulations, the model delivered orders-of-magnitude faster results, supporting rapid uncertainty quantification and scenario screening. This work departs from traditional scalar fracture predictions by directly modeling spatio-temporal geometry patterns from simulation data. The resulting surrogate offers an interpretable, computationally efficient tool that retains physics-informed accuracy, enabling real-time or near real-time fracture design optimization in unconventional reservoirs.

Introduction

Hydraulic fracturing has become a cornerstone of modern unconventional reservoir development, enabling the economic extraction of hydrocarbons from low-permeability formations. However, as operations scale to field-wide developments involving dozens or even hundreds of wells, the complexity of fracture modeling grows exponentially. Full-physics simulations—while accurate—are computationally expensive and time-consuming, making it infeasible to exhaustively model every treatment scenario across a field. In such environments, engineers must contend with vast heterogeneity in reservoir properties, high degrees of uncertainty, and tight operational timelines. This underscores a critical industry need: robust, fast, and generalizable surrogate models that maintain accuracy while enabling real-time or near real-time decision making.

Modeling fracture geometry across large fields is especially laborious due to the intricate coupling between geomechanical, fluid, and operational parameters. Rock deformation, fracture propagation, and fluid distribution evolve in space and time, requiring high-resolution numerical solvers and long runtimes. Consequently, operators often face a trade-off between model fidelity and computational efficiency—hindering their ability to run probabilistic analyses, evaluate development scenarios, or adjust in real time during fracturing treatments. A model that can retain physical insight while delivering predictions within seconds has the potential to transform fracture design workflows from reactive to proactive.

A key challenge in developing predictive models for hydraulic fracturing lies in integrating the physical properties of the reservoir (e.g., Young's modulus, Poisson's ratio, in-situ stress) with the dynamic nature of the fracture stimulation job. These interactions evolve nonlinearly with time and spatial distribution, requiring models that not only interpolate across known conditions but generalize well under uncertainty. Deep learning (DL) algorithms such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs) offer powerful capabilities for handling non-linear data and spatial distributions (Lu et al., 2022; Zhao et al., 2024). However, they come with notable trade-offs. Deep learning typically requires large labeled datasets, significant computational resources (e.g., GPUs), and longer training times. Additionally, DL models often operate as "black boxes," limiting interpretability, which is a drawback in engineering applications where transparency and physical reasoning are vital.

Extreme Gradient Boosting (XGBoost), on the other hand, is a powerful machine learning algorithm well-suited for structured tabular data and nonlinear relationships (Chenand Guestrin,2016). XGBoost has emerged as a leading tool for spatio-temporal prediction tasks in scientific computing due to its ability to handle feature interactions, regularize overfitting, and operate efficiently at scale without compromising interpretability (Niazkar et al., 2024). In petroleum engineering, it has been widely used toimprove assisted history matching workflows (Zeng et al., 2024), predict rate of penetration during well construction (Zhou et al., 2022), and prediction of porosity curves using petrophysical data (Pan et al., 2022). In hydraulic fracturing, XGBoost applications are limited, mostly due to continuous traditional scalar predictions (e.g., fracture length based on microseismic clouds), which deter the understanding of 3D fracture propagation. Among the limited work of XGBoost on hydraulic fracturing,it has been applied to predict planar hydraulic fracturing geometries (Yang et al., 2024). However, non-planarity may exist when fracture propagation interacts with rock discontinuities and encounters low stress contrast environments.

This paper introduces anew XGBoost-based framework that aims to solve that limitation by capturing the directional growth of the fracture front in both lateral and vertical dimensions. It achieves this by learning from representative 3D points extracted from high-fidelity fracture simulations—effectively mapping spatial propagation patterns across the entire treatment duration with no fracture front direction restrictions. XGBoostexcels at capturing these complex patterns without requiring extensive hyperparameter tuning or vast training datasets.It works efficiently on smaller datasets, and it supports rapid training and inference on standard CPUs.By incorporating physical insight from simulations into a surrogate model, this workflow allows engineers to quantify uncertainty, analyze development options, and optimize fracturing strategies in a fraction of the time required by full-physics models. Ultimately, the goal is to enable smarter, faster decisions in the field—whether for treatment design, real-time adjustments, or field development planning.

Methodology

Data preparation

To train a reliable surrogate model for hydraulic fracture geometry prediction, the proposed workflow begins by systematically sampling a wide range of input parameters that influence fracture propagation. These include geomechanical properties (e.g., Young's modulus, Poisson's ratio, minimum and maximum horizontal stresses, fracture toughness), fracturing operation parameters (e.g., injection rate, fluid viscosity, proppant concentration, proppant type), and completion design attributes (e.g., cluster spacing). Natural fracture characteristics (e.g., density and spatial distribution) are handled using a static model. While these inputs are standard in full-physics fracture simulations, they are repurposed here as features for the XGBoost model to capture directional growth trends in both the lateral and vertical directions. These input parameters span a high-dimensional design space. To efficiently explore this space while ensuring diverse and representative coverage, we employ Latin Hypercube Sampling (LHS). LHS is a stratified sampling technique well-suited for high-dimensional parameter studies. This method enables more efficient coverage of the input space with fewer samples, which is particularly valuable when dealing with computationally intensive simulations.

Once the input parameters have been sampled using Latin Hypercube Sampling (LHS), the next step involves running 100 hydraulic fracturing simulations, each corresponding to a unique set of input properties. Each simulation produces a distinct 3D fracture geometry that serves as the target output for training the XGBoost model. These simulations are designed to capture the complex, time-dependent behavior of hydraulic fractures, where the geometry evolves over time as the fracturing process unfolds. At every timestep during the simulation, the evolving fracture geometry is read and stored in an efficient data structure that tracks crucial attributes, such as fracture element pressure, fracture width, and proppant concentration. These properties are essential for accurately characterizing fracture growth in both the lateral and vertical directions.

Definition of 3D Stencils for Data Extraction

The fracture geometry is represented by a series of 3D stencils that capture the fracture characteristics at various stages of the hydraulic fracturing process. These stencils are not confined to a single plane, but rather span multiple planes, resulting in a non-planar fracture shape. This non-planar representation is essential for modeling fractures that exhibit complex geometries, such as branching or curved fractures, which are common in real-world scenarios with complex reservoir architectures.

For each stage of the hydraulic fracturing process, the 3D stencils are defined along the hydraulic fractures within the fracturing stage. The stencils are organized as a series of discrete points in the x-y plane (pink lines in Fig. 1), which are then extended into the z-direction to capture the full 3D geometry of the fracture (green dots in Fig.1). By doing this, we ensure that the model can account for the fracturing process in three dimensions, reflecting the actual spatial distribution of the fracture elements. The fracture evolution can be observed in detail in Fig. 1, which features a non-planar hydraulic fracture.

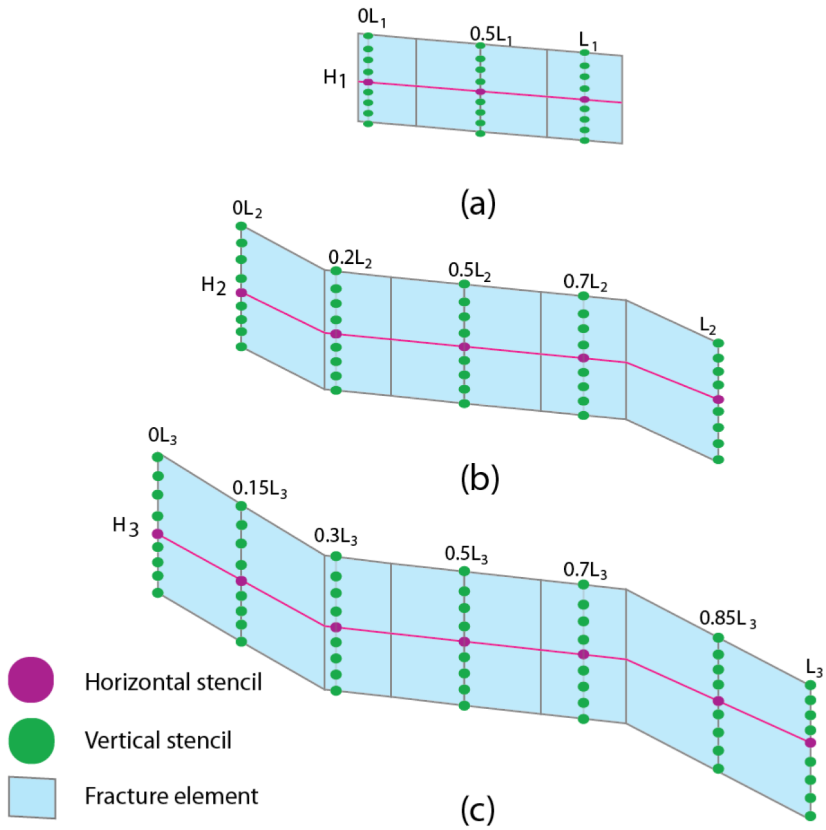

Figure 1—Non-planar hydraulic fracturing growth with horizontal (pink) and vertical (green) stencils; a) First timestep (length, L1; height H1), b) Second timestep (length, L2; height H2), c) Third timestep (length, L3; height H3).

Data Extraction from the Fracture Geometry

At each x-y-z point of the stencils, the fracture properties (such as width, pressure, and proppant concentration) are accessed. These properties are extracted for each simulation at each timestep, forming a comprehensive dataset of fracture behavior over time. The extracted data at each point serves as the input features for the XGBoost model.

This step is crucial for providing the model with high-dimensional data that reflects the spatial evolution of the fracture, including the pressure distribution, width variation, and proppant flow characteristics. Each point in the 3D stencil grid contributes to the training data used by the model, ensuring that the spatial and temporal dynamics of fracture propagation are captured in the predictive features.

Preparation for XGBoost Training

The data obtained from the 3D stencils—including fracture width, pressure, and proppant concentration at each x-y-z point—are organized into a structured format suitable for training the XGBoost model. This training dataset contains the essential features of fracture behavior at various stages, enabling the model to learn the underlying patterns of fracture propagation. The model then uses this data to predict fracture geometry under varying input conditions, including the lateral and vertical directions. By systematically combining the fracture properties extracted from the simulation results with the parameter space defined by LHS, we create a robust training dataset for the XGBoost model that accounts for both spatial complexity and uncertainty in reservoir conditions.

Results

We evaluated the performance of the XGBoost surrogate model using 3 hydraulic fracturing cases influenced by natural fractures. The simulated reservoir had a thickness of 20 meters, a burial depth of 3659 meters, and a leak-off coefficient of 0.0003 m/s$^{0.5}$. A total of 28 pumping stages were performed over 139 minutes, with a proppant density of 3200 kg/m^3. Natural fractures introduced curvature and asymmetry into the fracture geometry, providing a realistic test for the model. The training included 3 key geomechanical parameter ranges for each case and are summarized in Table 1.

Table 1—Results of prediction model

Property	Range	Unit
Young's Modulus	66.16 - 70.16	MPa
Poisson ratio	0.2 - 0.4	-
Minimum horizontal stress	102.4 - 107.55	MPa

Case 1

The surrogate model was trained using 561 spatial sampling points—20 horizontal and 8 vertical—extracted from the simulated fracture volume. Predictions were made every 200 seconds, covering 190-time steps. Once trained, the model could predict the full 3D fracture geometry in under 34 seconds. Training results are shown in Table 2. The full-physics simulation results show a large influence of natural fractures in the direction of the hydraulic fracture (Fig. 2). Compared to the full-physics simulator, the surrogate achieved a fracture length error of only 0.4% and a height error of 0.15%, demonstrating strong accuracy and robustness even with non-planar fractures. These results confirm that the model accurately reproduces spatial fracture propagation while offering significant computational speed-up. The results are summarized in Table 3 and Fig. 3.

Table 2—Results of prediction model

Property	Value	Unit
Training samples (spatial points)	561	-
Prediction frequency	200	s / timestep
Time steps predicted	190	-
Training time	4264	s
Full-physics simulation time	43	s
XGBoost prediction time	34	s

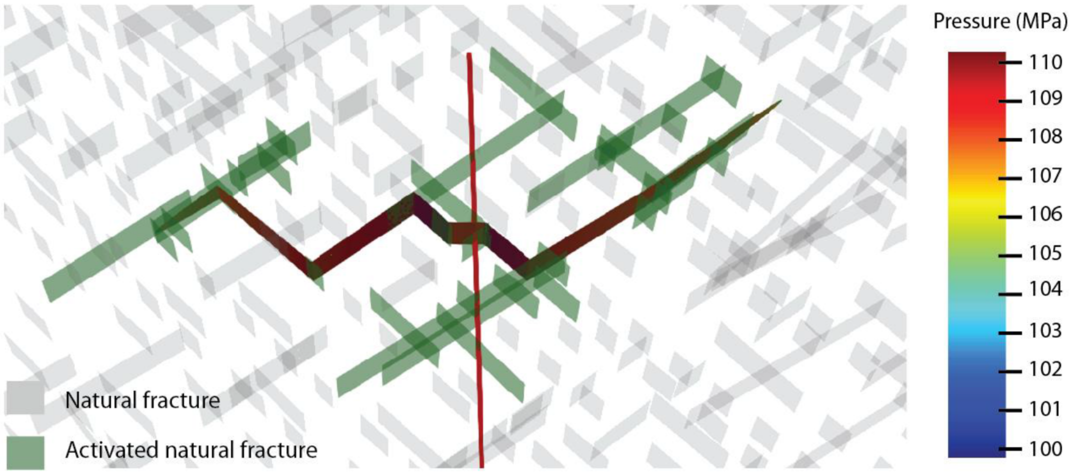

Figure 2—Non-planar hydraulic fracture geometry with one cluster in a naturally fractured reservoir

Table 3—Hydraulic fracturing geometry results of prediction model

Fracture	Fracture length (m)		Error (%)	Fracture height (m)		Error (%)
	Simulation	Prediction		Simulation	Prediction	
1	160.1	159.4	0.4	19.83	19.80	0.15

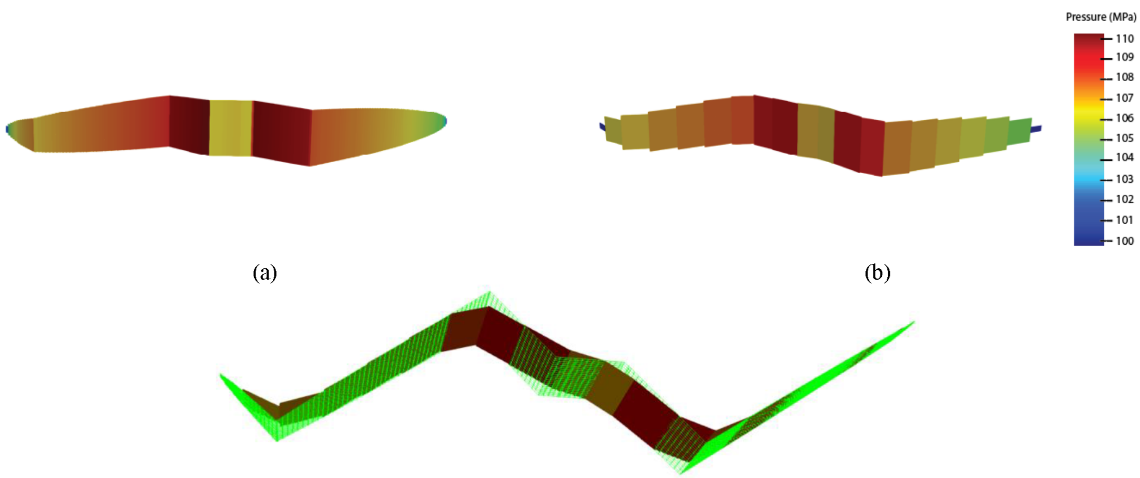

Figure 3—Hydraulic fracture results: a) hydraulic fracture simulation results; b) hydraulic fracture prediction results; c) fracture geometries overlap (green mesh is the simulation results and colored picture is the prediction)

Case 2

We tested the XGBoost surrogate model on a more complex scenario involving a single-stage hydraulic fracturing treatment with two clusters spaced 38 meters apart. A total of 1121 sampling points were extracted from the simulation, combining 20 horizontal and 8 vertical locations per cluster. The surrogate predicted fracture growth every 50 seconds across 170-time steps. The model required approximately 71 minutes to train and produced rapid predictions thereafter. Training results are outlined in Table 4. The stimulation followed the same 139-minute pumping schedule divided into 28 stages, with a proppant density of 3200 kg/m³. This multi-cluster setup introduced interactions between fractures, making it an effective test of the model's capacity to capture non-planar and stress shadow behavior. Simulation results are shown in Fig. 4.

Table 4—Results of prediction model

Property	Value	Unit
Training samples (spatial points)	561 × 2=1121	-
Prediction frequency	50	s / timestep
Time steps predicted	170	-
Training time	4264	s
Full-physics simulation time	91	s
XGBoost prediction time	43	s

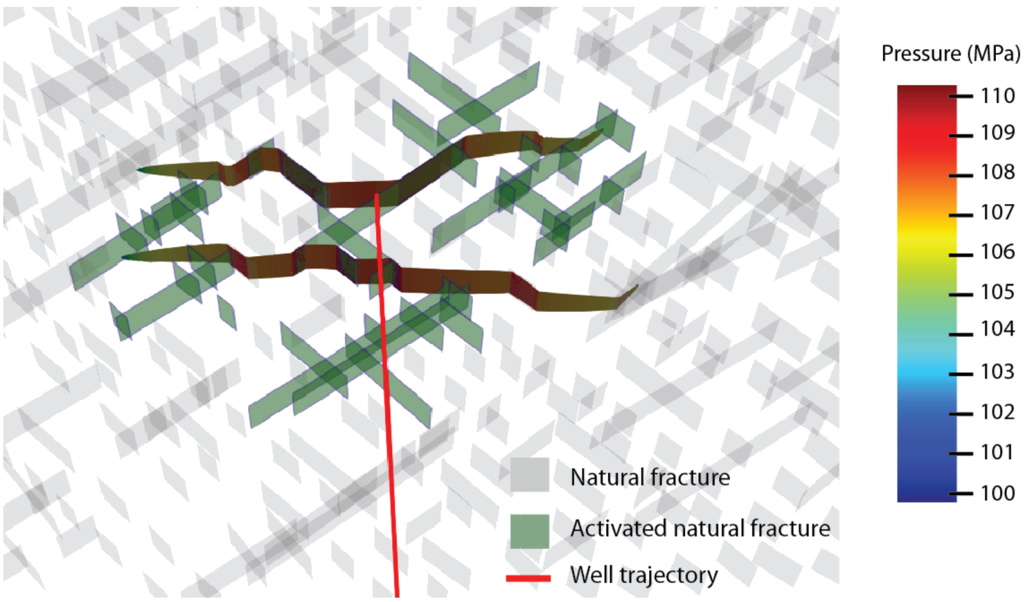

Figure 4—Non-planar hydraulic fracture geometry with two clusters in a naturally fractured reservoir

Despite the added complexity of cluster interactions, the model maintained high accuracy, preserving spatial growth patterns and asymmetries. Average error in length and height for the 2 fractures is 0.73% and 1.84%, respectively. These results show that the surrogate model efficiently handles multi-cluster scenarios with interacting fractures, preserving geometry fidelity while significantly reducing simulation time. The hydraulic fracturing geometry prediction results are outlined in Table 5 and Fig. 5, respectively.

Table 5—Hydraulic fracturing geometry results of prediction model

Fracture	Fracture length (m)		Error (%)	Fracture height (m)		Error (%)
	Simulation	Prediction		Simulation	Prediction	
1	212.767	210.016	1.29%	19.183	19.536	1.84%
2	218.119	217.757	0.17%	19.183	19.536	1.84%

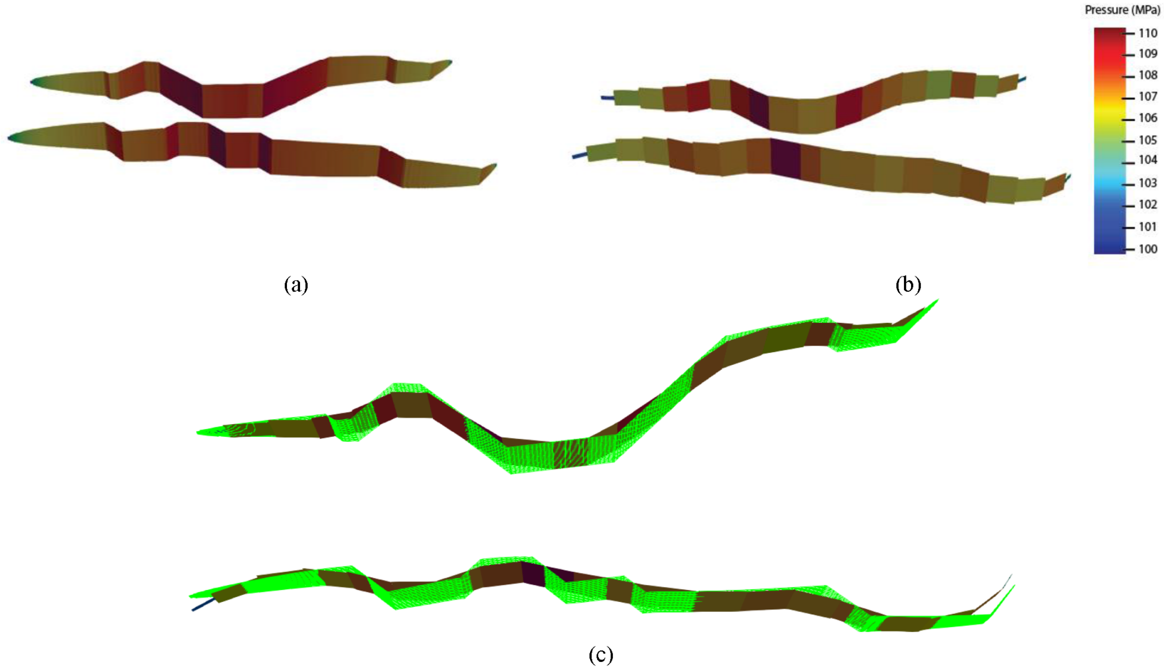

Figure 5—Hydraulic fracture results: a) hydraulic fracture simulation results; b) hydraulic fracture prediction results; c) fracture geometries overlap (green mesh is the simulation results and colored picture is the prediction)

Case 3

We evaluated the XGBoost surrogate model on a complex curved multi-fracture scenario involving three clusters per stage, each spaced 45 meters apart. A total of 1683 sampling points were extracted from the simulation, combining 20 horizontal and 8 vertical positions per cluster. Predictions were generated every 50 seconds over 170-time steps and are summarized in Table 6. The model required 6587 seconds for training, with the full-physics simulation completing in 149 seconds (Fig. 6) and surrogate predictions in 32 seconds. The results preserved complex non-planar growth patterns and inter-cluster interactions while providing significant computational speed-up.

Table 6—Results of prediction model

Property	Value	Unit
Training samples (spatial points)	561 × 3=1683	-
Prediction frequency	50	s / timestep
Time steps predicted	170	-
Training time	6587	s
Full-physics simulation time	149	s
XGBoost prediction time	32	s

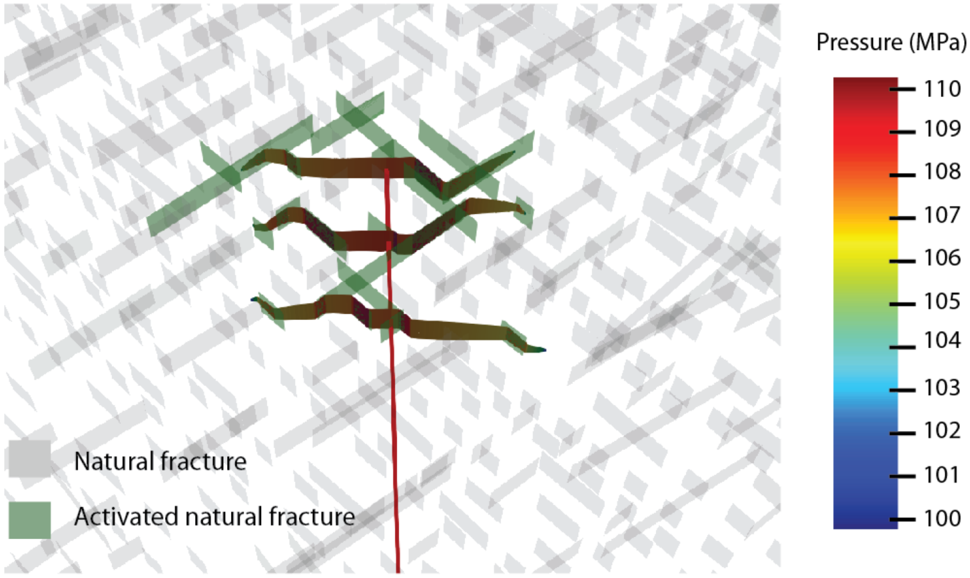

Figure 6—Non-planar hydraulic fracture geometry with two clusters in a naturally fractured reservoir

Average length and height error corresponds to 1.05% and 1.4%, respectively. These findings confirm that the surrogate model maintains high accuracy in predicting spatially complex fracture networks under natural fracture influence while delivering rapid predictions suitable for uncertainty analysis and design optimization. The results are outlined in Table 7 and Fig. 7.

Table 7—Hydraulic fracturing geometry results of prediction model

Fracture	Fracture length (m)		Error (%)	Fracture height (m)		Error (%)
	Simulation	Prediction		Simulation	Prediction	
1	145.566	144.023	1.6%	19.812	19.535	1.4%
2	141.47	140.132	0.95%	19.812	19.535	1.4%
3	142.572	143.416	0.59%	19.812	19.535	1.4%

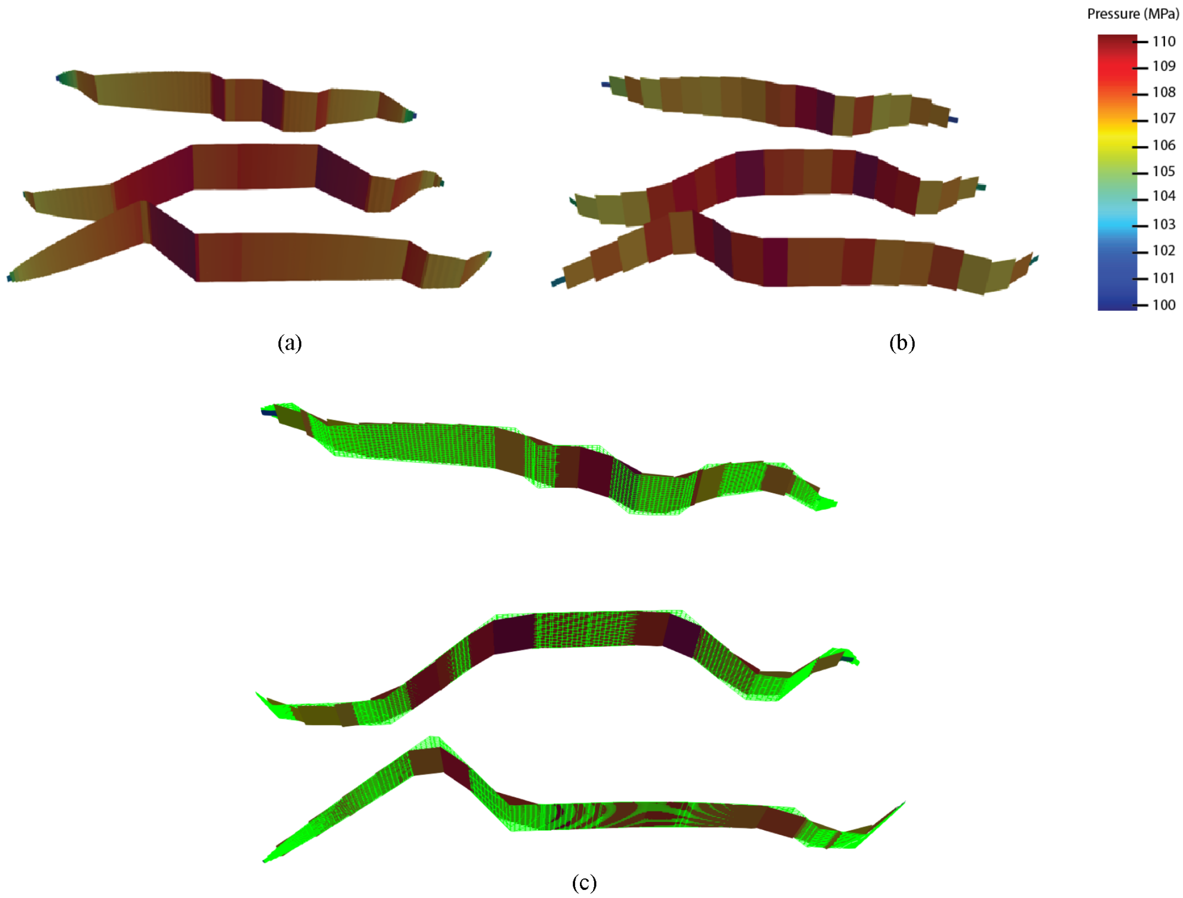

Figure 7—Hydraulic fracture results: a) hydraulic fracture simulation results; b) hydraulic fracture prediction results; c) fracture geometries overlap (green mesh is the simulation results and colored picture is the prediction)

Conclusions

This study demonstrated that an XGBoost-based surrogate model can deliver fast and accurate predictions of 3D hydraulic fracture geometries under a wide range of reservoir and operational conditions. By combining Latin Hypercube Sampling with structured 3D stencils, the workflow captured the spatio-temporal evolution of fracture propagation, including the influence of natural fractures and cluster interactions. Across single- and multi-cluster scenarios, the model reproduced fracture length and height with errors below 1%, while also preserving curvature and asymmetry, confirming its ability to generalize beyond planar assumptions.

One of the main strengths of this approach lies in its efficiency and interpretability. Once trained, the model can predict entire fracture time series in seconds to minutes, providing speed-ups of several orders of magnitude compared to repeated full-physics simulations. Its ability to highlight feature importance offers engineers direct insight into which geomechanical and operational parameters most strongly govern fracture growth, an advantage over more opaque deep learning models. Additionally, XGBoost requires relatively modest training datasets, making it practical for projects where simulation data are limited.

At the same time, the approach has inherent limitations. An initial investment of physics-based simulations is still required, which may be costly for projects lacking existing simulation infrastructure. While XGBoost performs well with tabular and structured data, it may struggle to capture highly complex spatial correlations compared to deep learning methods designed for image-like or sequential data. Furthermore, model performance depends on consistent stencil definitions, which may reduce accuracy if fracture growth deviates strongly from the expected patterns. Extending the framework to multi-stage, field-scale scenarios will also require further validation.

Overall, the results confirm that XGBoost offers a balanced compromise between accuracy, speed, and interpretability. It represents a practical tool for accelerating fracture design, enabling rapid uncertainty quantification, scenario screening, and near real-time decision making in unconventional field development. Future work should focus on scaling the workflow to larger multi-stage systems and exploring hybrid approaches that combine gradient boosting with deep learning to capture more detailed spatial dynamics.

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